

EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** Seoul, Korea
Ph.D. Student - AI Graduate School - HuGe Lab - Advisor: Beomjoon Kim *Sep 2022 - Present*
- **Seoul National University** Seoul, Korea
M.S. in Mechanical Engineering - Advisor: Dongjun Lee *March 2018 - Feb 2020*
Thesis: Multi-Contact Simulator and Reinforcement Learning for Screw Tightening Tasks
- **Seoul National University** Seoul, Korea
B.S. in Mechanical Engineering *March 2010 - Feb 2014*

EXPERIENCE

- **Allen Institute for AI (Ai2)** Seattle, WA
Research Intern, Robotics Group *Oct 2025 - June 2026*
 - **Torque Adaptation:** Worked on a torque adaptation module to reduce the sim-to-real gap under the mentorship of Dieter Fox.
- **Samsung Research** Seoul, Korea
AI Researcher - AI Methods Team *Feb 2020 - June 2022*
 - **Robot Learning:** Developed robot learning algorithms for vision-based manipulation, including cluttered-scene grasping, sim-to-real transfer, planning, and data-efficient reinforcement learning.
- **Samsung/Hanwha Techwin** Changwon, Korea
QA Engineer - Gas Turbine Engine Division *May 2014 - Sep 2017*

PUBLICATIONS

- **CODA: Depth-Aligned Scene Completion and Object Decomposition from a Single RGB-D Image:** Dongwon Son, Junhyek Han, Yoontae Cho, Minseok Lee, Hong-seok Choi, Jiwook Choi, Hyungjin Kim, and Beomjoon Kim. arXiv preprint arXiv:2609.25654. 2026. paper site
- **TAM: Torque Adaptation Module for Robust Motion Transfer in Manipulation:** Dongwon Son, Florian Shkurti, Jason Lee, Naman Shah, Beomjoon Kim, and Dieter Fox. CoRL. 2026. paper site
- **NeuralSVCD for Efficient Swept Volume Collision Detection:** Dongwon Son, Hojin Jung, and Beomjoon Kim. CoRL. 2025. paper site
- **DEF-oriCORN: efficient 3D scene understanding for robust language-directed pick-and-place:** Dongwon Son, Sanghyeon Son, Jaehyung Kim, Hojin Jung, and Beomjoon Kim. under review. 2025. site
- **An Intuitive Multi-Frequency Feature Representation for SO(3)-Equivariant Networks:** Dongwon Son, Jaehyung Kim, Sanghyeon Son, and Beomjoon Kim. ICLR. 2024. paper site
- **Preference learning for guiding the tree search in continuous POMDPs:** Jiyong Ahn, Sanghyeon Son, Dongryung Lee, Jisu Han, Dongwon Son, and Beomjoon Kim. CoRL. 2023. video paper site
- **Local Object Crop Collision Network for Efficient Simulation of Non-Convex Objects in GPU-based Simulators:** Dongwon Son, and Beomjoon Kim. R:SS. 2023. video paper site
- **Grasping as Inference: Reactive Grasping in Heavily Cluttered Environment:** Dongwon Son. RA-L. 2022. video paper

- **Reinforcement Learning for Vision-based Object Manipulation with Non-parametric Policy and Action Primitives:** Dongwon Son, Myungsin Kim, Jaechol Sim, and Wonsik Shin. IROS. 2021. video paper
- **Sim-to-Real Transfer of Bolting Tasks with Tight Tolerance:** Dongwon Son, Hyunsoo Yang, and Dongjun Lee. IROS. 2020. video paper
- **Learnable Environment Model with Data Efficiency for MPC of Assembly Tasks:** Dongwon Son, Hyunsoo Yang, and Dongjun Lee. IROS Workshop LRPC. 2019. video paper
- **Data-driven Contact Clustering for Robot Simulation:** Myungsin Kim, Jaemin Yoon, Dongwon Son, and Dongjun Lee. ICRA. 2019. paper

SKILLS SUMMARY

- **Languages:** Python, C++, MATLAB
- **Frameworks:** TensorFlow, Keras, JAX, ROS, OpenGL, PyBullet, Open3D, OMPL, FCL
- **Tools:** SolidWorks, EasyEDA

ENGINEERING PROJECTS

- **Manipulator Identification:** Identified physics parameters for Franka Emika Panda.
- **Manipulator Controller Design:** Implemented impedance, admittance, and compliance controllers.
- **Motor Driver Design and FOC Control:** Built PCB and MCU control stack with anticogging and FOC control.

RESEARCH INTERESTS

- Robot manipulation, including object manipulation, grasping, vision-based assembly, and long-horizon manipulation
- Computationally efficient simulation, including physics engines, collision detection, and rendering
- Sim-to-real transfer
- Structured network design and shape representation for manipulation and planning
- Open-source, cost-efficient manipulator design